PHYSIOLOGY

Liver Physiology

Physiology GIT Block · MEDucation by Mattin Majid

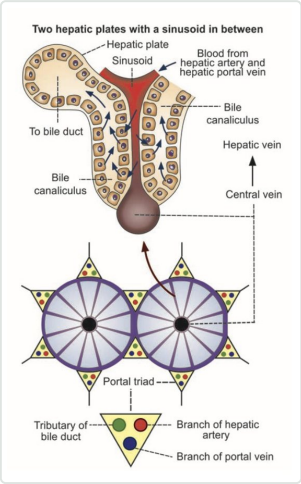
BEFORE YOU READ — ANSWER FROM MEMORY

1. Two vessels bring blood to the liver — what share does each carry, and at what pressure?
2. Why does a rise of only a few mmHg in hepatic vascular pressure produce ascites?
3. What does CCK do to the gallbladder and the sphincter of Oddi, and what does secretin add to bile?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The lobule — the functional unit

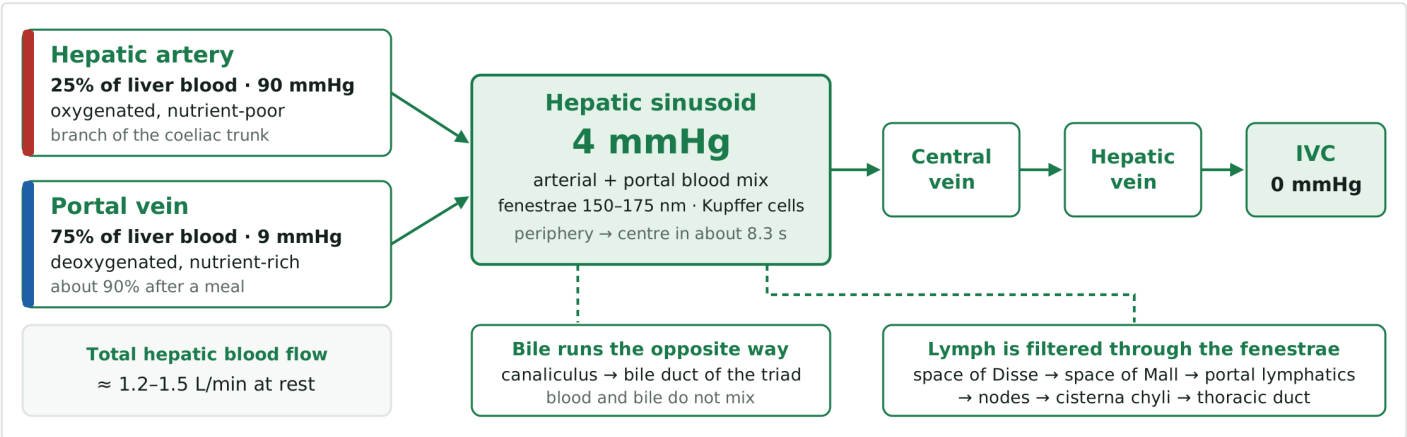
- **Largest internal organ and gland**. Two principal lobes — **large right, smaller left**; by blood-vessel distribution the left also includes the **quadrate and caudate** lobes.
- **Lobules are the functional units** — 50,000 to 100,000 per liver, each hexagonal (honeycomb-like).
- **Central vein in the middle; portal triads at the periphery** — branches of the hepatic artery, portal vein and bile duct, opening into the sinusoids.
- **Hepatocytes** (up to **30% double-nucleated**) form **plates 1–2 cells thick**; a **bile canaliculus** lies between the two columns of each plate.
- **Sinusoids** are the large capillary spaces between plates; their endothelium has **fenestrae 150–175 nm** wide, making them very porous.
- **Kupffer cells** in the sinusoids are phagocytic: **almost no gut bacteria** pass from portal blood into the systemic circulation.
- **Cholangiocytes** (polarized epithelium) line the bile channels; they **modify and transport bile**.
- Blood flows **periphery → central vein** (about **8.3 seconds**); central veins join to form the **hepatic vein → inferior vena cava**.



Hepatic plates and portal triad.

2 Dual blood supply and the pressure staircase

VISUAL ANCHOR · TWO INFLOWS, ONE LOW-PRESSURE SINUSOID



	Hepatic artery	Portal vein
Share of supply	**25%** of liver blood	**75%** of liver blood — **90% after a meal**
Comes from	Branch of the **coeliac trunk** (from the abdominal aorta)	**Mesenteric and splenic veins** — drains stomach, intestine, spleen and pancreas
Blood carried	**Oxygenated, nutrient-poor**; smaller volume, higher oxygen concentration	**Deoxygenated, nutrient-rich**: monosaccharides, amino acids; also bile salts, bilirubin, urobilinogen, GI hormones. Still a **substantial contribution to hepatic oxygen delivery**
Pressure	**90 mmHg** (high)	**9 mmHg** (low)

Slide 11 says 25% “of cardiac output”; slides 12, 15 and 17 give 25% / 75% as shares of liver blood — used here.

- Hepatocytes take **oxygen, most nutrients and certain toxic substances** from the sinusoid; what they manufacture is secreted back into the **central veins → hepatic veins → IVC**.
- Not everything absorbed is usable as absorbed: the liver **changes it to more usable forms** and **eliminates waste products and ingested toxins**.

HEPATIC ARTERIAL BUFFER RESPONSE

Portal flow falls — Less blood arrives through the portal vein → **Less washout** — Locally produced **adenosine** is cleared more slowly → **Adenosine builds up** — Hepatic **arterioles dilate**; arterial resistance falls → **Arterial flow rises** — **Partial** compensation only

Limits: it helps **stabilise hepatic oxygen delivery** but does not completely preserve total liver blood flow, and cannot be assumed to protect the biliary tree from loss of its arterial supply.

3 Why low pressure matters — ascites

Cue	Detail
Why pressure must be low	(1) To **prevent massive fluid leakage** — plasma proteins cross the large endothelial openings freely, so the protective osmotic pressure is eliminated and **hydrostatic pressure alone** controls fluid movement; (2) to maximise nutrient and oxygen exchange
Trigger	Hepatic vascular pressure only **3–7 mmHg above normal** → fluid transudes into lymph and leaks through the liver capsule into the abdominal cavity
The fluid	**Almost pure plasma** — 80–90% as much protein as normal plasma
Vena caval pressure 10–15 mmHg	Hepatic **lymph flow up to 20 times normal**; “sweating” from the liver surface → large amounts of free fluid = **ascites**
Portal flow blocked	High capillary pressure in the **entire portal system** → oedema of the gut wall → transudation through the gut serosa → also ascites
Also on the slide	Decreased hepatic albumin synthesis → hypoalbuminaemia → **decreased plasma oncotic pressure**

4 Blood reservoir and lymph

Cue	Detail
Blood reservoir	Large, expandable venous organ holding about **10% of total blood volume** — stores blood when volume is in excess, supplies it when volume falls
High right atrial pressure	In **cardiac failure**, an extra **0.5–1 litre** is stored in the hepatic veins and sinuses
Lymph output	**Largest lymph-producing organ** — 25–50% of the lymph passing through the thoracic duct
Where lymph forms	Plasma filtered through the fenestrae of the sinusoidal endothelial cells (LSECs) enters the **space of Disse**, between LSECs and hepatocytes (**hepatic stellate cells** sit here). It then passes through the **space of Mall** — between the portal-tract stroma and the outermost hepatocytes — into the **portal-tract lymphatics**, where most lymphatic vessels lie
Immune role	Antigen-presenting cells (dendritic cells, macrophages) in lymph meet lymphocytes in the draining nodes → **adaptive immune responses**

LYMPH PATHWAY (DEEP SYSTEM)

Sinusoid — Filtration through fenestrae → **Space of Disse** — then **space of Mall** → **Portal lymphatics** — converge on the porta hepatis → **Lymph nodes** — hepatic, regional; coeliac and para-aortic pathways → **Cisterna chyli** — then **thoracic duct** → **Left venous angle** — left internal jugular + left subclavian

5 Bile — what is in it and how it is made

- About **0.5–1 L of bile per day**; production **begins in hepatocytes** and continues with **modification by cholangiocytes**.
- Major constituents: **bile pigment (bilirubin), bile salts, phospholipids (mainly lecithin), cholesterol and inorganic ions**.

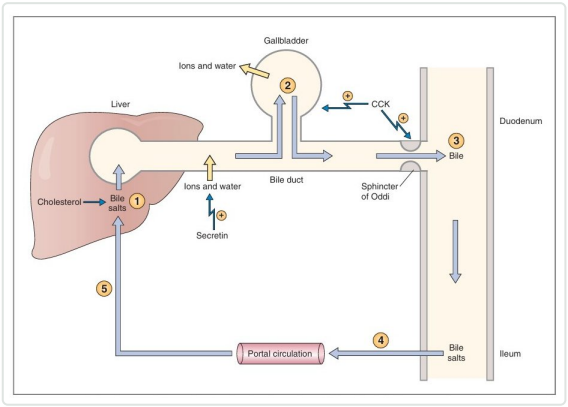
Cue	Detail
Bilirubin	The principal bile pigment. Phagocytosis of aged red cells liberates iron, globin and bilirubin (from heme); **iron and globin are recycled**, bilirubin goes into bile. Its breakdown product **stercobilin gives faeces their brown colour**
Bile salts	Sodium and potassium salts of bile acids (mostly **chenodeoxycholic and cholic acid**). **Emulsify** large lipid globules into small ones → large surface area → **pancreatic lipase** digests triglycerides faster
Ductules reclaim	**Glucose and amino acids** returned to the circulation by active transport — likely to **prevent bacterial overgrowth in bile**, especially during gallbladder storage
Ductules also	Hydrolyse glutathione to its amino acids with **GGT** (gamma-glutamyltranspeptidase, on the apical membrane of cholangiocytes); secrete **bicarbonate in response to secretin** (post-prandial), plus **IgA and mucus** for protection

ROUTE OF BILE

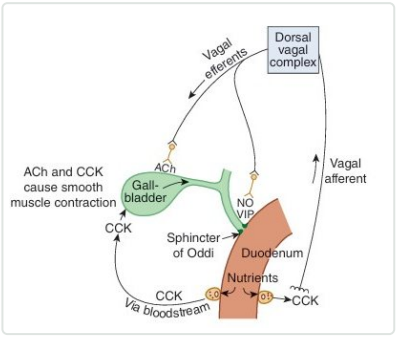
Bile canaliculi within each hepatic plate → **Interlobular ducts** drain the canaliculi → **R + L hepatic ducts** join outside the liver → **Common hepatic duct** joined by the **cystic duct** → **Common bile duct** → **Ampulla of Vater** opens into the **duodenum**

6 Storage, release and recycling of bile

- Bile leaves the liver, fills the **gallbladder** and is stored; the gallbladder **concentrates it by absorbing water and ions**.
- In the small intestine bile salts **emulsify and solubilise lipids**.
- After lipid absorption, bile salts return **from the ileum through the portal system** and are extracted by hepatocytes.
- This recirculation **reduces the need to synthesise new bile salts**.



Enterohepatic circulation of bile salts: (1) secretion, (2) storage and concentration, (3) release, (4) ileal uptake, (5) return in portal blood.



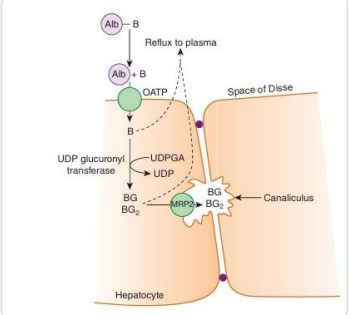
CCK contracts the gallbladder and triggers a vagovagal reflex (ACh); the sphincter of Oddi relaxes (NO, VIP).

Signal	Released / triggered by	Effect on the biliary system
CCK	**Intestinal I cells**, when chyme brings digestion products — especially **fats and proteins** (fatty acids, amino acids) — into the duodenum	**Gallbladder contraction + relaxation of the sphincter of Oddi** → stored bile enters the duodenum. Also regulates pancreatic enzyme secretion and GI motor activity
Secretin	Post-prandial period	Increases the **water and HCO_3^- content** of bile
Vagus nerves	Vagal stimulation	**Increases bile production**
Food in mouth	Neural and hormonal influences	**Resistance of the sphincter of Oddi falls**
Choleretics	Substances that increase bile secretion	**Bile acids themselves** are among the most important physiological choleretics

7 Bilirubin — from blood to bile

In blood Formed from haemoglobin breakdown; travels **bound to albumin** → **Uptake** Free bilirubin enters via **OATP**; binds cytoplasmic proteins → **Conjugation** **UDP-glucuronosyl-transferase** in the **smooth ER** → **Secretion** **MRP-2** pumps it into the canaliculus

- Each bilirubin reacts with **two UDPGA molecules → bilirubin diglucuronide**, which is **more water-soluble** than free bilirubin.
- Canalicular secretion runs **against a concentration gradient** — an active transporter (multidrug resistance protein-2).



Bilirubin handling by the hepatocyte.

8 The function list

Function	What the lecture says	Function	What the lecture says
Plasma proteins	Acute-phase, hormone-binding, clotting and complement proteins — **everything except immunoglobulins**	**Ammonia**	**Toxic to the CNS**, crosses the blood–brain barrier freely. Liver is the **only organ with the complete urea cycle** (Krebs–Henseleit) → urea → urine. Sources: mainly **colon and kidneys**; less from red-cell breakdown and muscle
Detoxification	Physical: **Kupffer cells** trap bacteria and particulates. Biochemical: **phase I** (oxidation, hydroxylation — cytochrome P450) then **phase II** (esterification) → inactive, less lipophilic metabolites	**Glucose**	Gluconeogenesis. The lecture names the liver as the **only organ with glucose-6-phosphatase**, so only it releases glucose from glycogen into blood; muscle uses its glycogen locally
Storage	Glycogen, amino acids, **iron, folic acid, vitamins A, B12 and D**	**Synthesis (other)**	Steroids, **somatomedin** and **heparin**
Excretion in bile	Cholesterol, bile pigments, **heavy metals** (lead, arsenic, bismuth), toxins, bacteria and viruses (e.g. yellow fever)	**Metabolism, heat**	Organ of **maximum metabolic reactions** (carbohydrates, proteins, fats, vitamins, many hormones) and of **maximum heat production**
Haemopoietic	Makes blood cells in the **fetus** (hepatic stage); stores B12 and iron; produces **thrombopoietin**	**Haemolytic**	Senile red cells (lifespan **120 days**) destroyed by Kupffer cells
Inactivation	Catabolises **GH, parathormone, cortisol, insulin, glucagon, oestrogen**; turns fat-soluble drugs into water-soluble ones excreted in bile or urine	**Defence**	Kupffer cells **phagocytose** bacteria and antigens and release **interleukins and tumour necrosis factors**

P Must-remember pearls

- **25% hepatic artery (90 mmHg) + 75% portal vein (9 mmHg) → sinusoid 4 mmHg → vena cava 0 mmHg.** Portal share rises to 90% after a meal.
- Sinusoids are so porous that proteins pass freely, so **hydrostatic pressure is the only force**: **3–7 mmHg above normal → ascites**, a fluid with 80–90% of plasma protein.
- **Hepatic arterial buffer response**: portal flow ↓ → adenosine accumulates → arterioles dilate. Compensation is **partial**.
- **CCK** (I cells; fats and proteins) = **gallbladder contracts + sphincter of Oddi relaxes**. **Secretin** = more water and HCO_3^-. Bile salts recycle **ileum → portal blood → liver**.
- Bilirubin: **albumin → OATP → UDP-glucuronosyltransferase (smooth ER) → MRP-2 → canaliculus**.
- The liver makes every major plasma protein **except immunoglobulins** and is the only organ with the **complete urea cycle**.

Q Test yourself

1. A 64-year-old man with right-sided heart failure has a vena caval pressure of 12 mmHg and a tense, fluid-filled abdomen. Which description of the fluid fits the mechanism in the lecture?

A. Protein-free fluid, because sinusoidal endothelium blocks plasma proteins

B. Bile-stained fluid leaking from distended bile canaliculi

C. Blood, from rupture of sinusoids at high arterial pressure

D. Almost pure plasma, with 80–90% as much protein as normal plasma

E. Fluid secreted by Kupffer cells in response to gut bacteria

2. After a fatty meal, a healthy woman’s gallbladder empties. Which signal best explains this?

A. Secretin from the duodenum, which contracts the gallbladder wall
B. CCK from intestinal I cells, contracting the gallbladder and relaxing the sphincter of Oddi
C. Bile acids returning in portal blood, which close the sphincter of Oddi
D. Adenosine accumulating in the liver, which dilates the cystic duct
E. CCK from hepatocytes, which increases the bicarbonate content of bile

3. A patient’s portal venous inflow suddenly drops. Within the liver, hepatic arterial flow then rises. What is the major explanation given for this?

A. Reduced washout lets locally produced adenosine accumulate and dilate hepatic arterioles
B. Kupffer cells release interleukins that constrict the portal venules
C. CCK released from I cells dilates the hepatic artery
D. Sinusoidal pressure rises to 90 mmHg and forces the arterioles open
E. Secretin increases hepatic arterial bicarbonate delivery

4. A teenager has a defect in the transporter MRP-2. Which step of bilirubin handling fails?

A. Binding of bilirubin to albumin in the circulation
B. Entry of free bilirubin into the hepatocyte
C. Conjugation of bilirubin with glucuronic acid in the smooth ER
D. Conversion of bilirubin to stercobilin in the intestine
E. Transport of bilirubin diglucuronide into the bile canaliculus

5. A fat-soluble drug is first hydroxylated by a cytochrome P450 enzyme and then esterified in hepatocytes. How does the lecture classify these two steps, and what is the result?

A. Phase II then phase I; an active, more lipophilic metabolite
B. Both are phase I; the drug is stored unchanged in the liver
C. Phase I then phase II; an inactive, less lipophilic metabolite
D. Both are phase II; the drug is trapped by Kupffer cells
E. Phase I then phase II; a metabolite that can only leave through lymph

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — D — High hepatic vascular pressure makes near-plasma fluid (80–90% of plasma protein) transude; proteins cross the fenestrae freely.

2 — B — Fats and proteins release CCK from I cells: gallbladder contracts, sphincter of Oddi relaxes.

3 — A — Hepatic arterial buffer response: less adenosine washout → arterioles dilate (partial compensation).

4 — E — MRP-2 actively moves bilirubin diglucuronide into the canaliculus, against a concentration gradient.

5 — C — Phase I = oxidation/hydroxylation by cytochrome P450; phase II = esterification → inactive, less lipophilic metabolites.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam